

A. Corpus construction (Sec. 4)

MP3Quran recordings
3 riwayaths $\times$ 5 reciters
Surahs 78--114

Forced alignment
(2 procedures)
manual reconciliation

7,998 ayah clips
11.17 h $\cdot$ mono 16 kHz
JSON timestamps

Quality control
recording / boundary /
integrity checks

B. Tajweed annotation (Sec. 5)

Rule specification
(Tajweed references)

Automatic candidate
extraction (Python)

Manual acoustic
boundary annotation

Expert verification
+ adjudication

C. Analysis, modelling and evaluation (Secs. 6--9)

8 acoustic
descriptors (Sec. 6)

5-class task +
imbalance handling

SVM $\cdot$ RF $\cdot$ KNN
XGBoost $\cdot$ MLP

Ablation, MFCC baseline,
cross-riwayah, bootstrap